

Assignment - I

NUMBER SYSTEMS

1) Convert the following with equivalents in binary, octal, and Hexadecimal.

i) 143.75 (decimal)

⇒ Binary

$$\begin{array}{r}
 2 \overline{) 143} \\
 2 \overline{) 71} - 1 \\
 2 \overline{) 35} - 1 \\
 2 \overline{) 17} - 1 \\
 2 \overline{) 8} - 1 \\
 2 \overline{) 4} - 0 \\
 2 \overline{) 2} - 0 \\
 1 - 0
 \end{array}$$

$$\therefore 143.75_{(10)} = 10001111.11_{(2)}$$

$$0.75 \times 2 = 1.5 \rightarrow 1$$

$$0.5 \times 2 = 1 \rightarrow 1$$

⇒ Octal

$$\begin{array}{r}
 8 \overline{) 143} \\
 8 \overline{) 17} - 1 \\
 8 \overline{) 2} - 1 \\
 0 - 2
 \end{array}$$

$$0.75 \times 8 = 6.00 \rightarrow 6$$

$$\therefore 143.75_{(10)} = 217.6_{(8)}$$

⇒ Hexadecimal

$$\begin{array}{r}
 16 \overline{) 143} - 15 (F) \\
 16 \overline{) 8} - 8 \\
 0
 \end{array}$$

$$0.75 \times 16 = 12.00 \rightarrow 12 \rightarrow C$$

$$\therefore 143.75_{(10)} = 8F.C$$

ii) $11011.101_{(2)}$

→ Decimal

$$1 \times 2^0 = 1$$

$$1 \times 2^1 = 2$$

$$0 \times 2^2 = 0$$

$$1 \times 2^3 = 8$$

$$1 \times 2^4 = 16$$

$$\underline{27}$$

$$1 \times 2^{-1} = 0.5$$

$$0 \times 2^{-2} = 0$$

$$1 \times 2^{-3} = 0.125$$

$$\underline{0.625}$$

$$\therefore 11011.101_{(2)} = \underline{27.625_{(10)}}$$

→ Octal

$$\underline{011011}$$

$$3 \quad 3$$

$$\underline{101}$$

$$5$$

$$\therefore 11011.101_{(2)} = \underline{33.5_{(8)}}$$

→ Hexadecimal

$$\underline{00011011}$$

$$1 \quad 11 \rightarrow (B)$$

$$\underline{01010}$$

$$5$$

$$\therefore 11011.101_{(2)} = \underline{1B.A_{(16)}}$$

iii) $453_{(8)}$

→ Decimal

$$3 \times 8^0 = 3$$

$$5 \times 8^1 = 40$$

$$4 \times 8^2 = 256$$

$$\underline{299}$$

$$\therefore 453_{(8)} = \underline{299_{(10)}}$$

→ Binary

$$4 \quad 5 \quad 3$$

$$100 \quad 101 \quad 011$$

$$\therefore 453_{(8)} = \underline{100101011_{(2)}}$$

→ Hexadecimal

$$= \underbrace{100101011}_{\substack{1 \quad 2 \quad B}} \therefore 453(8) = 12B(16)$$

iv) 1F.C (16)

→ Binary

$$1 - 0001$$

$$F - 1111$$

$$C - 1100$$

$$1F.C(16) = 11111.11(2)$$

→ Decimal

$$F \times 16^0 = F(15)$$

$$1 \times 16^1 = 16$$

$$31$$

$$C \times 16^{-1} = 0.75$$

$$\therefore 1F.C(16) = 31.75(10)$$

→ Octal

$$011111$$

$$3 \quad 7$$

$$110$$

$$3$$

$$\therefore 37.6$$

$$\therefore 1F.C(16) = 37.3(8)$$

Decimal

$$143.75$$

$$27.625$$

$$299$$

$$31.75$$

Binary

$$10001111.11$$

$$11011.101$$

$$100101011$$

$$11111.11$$

Octal

$$217.6$$

$$33.5$$

$$453$$

$$37.3$$

Hexa

$$8F.C$$

$$1B.A$$

$$12B$$

$$1F.C$$

2) Find equivalents

i) $324_5 = ?(10)$

$$4 \times 5^0 = 4$$

$$2 \times 5^1 = 10$$

$$3 \times 5^2 = 75$$

$$\underline{89}(10)$$

iii) $231_{10} = ?(7)$

$$7 \overline{) 231}$$

$$7 \overline{) 33} - 0$$

$$4 \overline{) 5} - 5$$

$$\therefore 231(10) = 450(7)$$

ii) $32 \cdot 23_{(4)} = ?_{10}$

$$2 \times 4^0 = 2$$

$$3 \times 4^1 = 12$$

$$\underline{14}$$

$$\therefore 32 \cdot 23_{(4)} = \underline{14 \cdot 6875}_{(10)}$$

$$2 \times 4^{-1} = 0.5$$

$$3 \times 4^{-2} = 0.1875$$

$$\underline{0.6875}$$

iv) $189_{(10)} = ?_6$

$$6 \overline{) 189}$$

$$6 \overline{) 31} - 3$$

$$6 \overline{) 5} - 1$$

$$0 - 5$$

$$189_{(10)} = 513_6$$

3) Find the radix x :

a) $(312/20)_x = (13.1)_x$

$$312_x = 3x^2 + x + 2$$

$$20 = 2x$$

$$13 = 1x + 3 + 1/x$$

$$\Rightarrow \frac{3x^2 + x + 2}{2x} = 1x + 3 + \frac{1}{x}$$

$$3x^2 + x + 2 = 2x^2 + 6x + \frac{2x}{x}$$

$$3x^2 - 2x^2 + x - 6x - 2x = 0$$

$$x^2 - 5x = 0$$

$$x(x-5) = 0$$

$$\Rightarrow x=5, x=0 \therefore x=5$$

$$b) 264_r + 136_r = 433_r$$

$$264 = 2r^2 + 6r + 4$$

$$136 = 1r^2 + 3r + 6$$

$$433 = 4r^2 + 3r + 3$$

$$2r^2 + 6r + 4 + r^2 + 3r + 6 = 4r^2 + 3r + 3$$

$$3r^2 + 9r + 10 = 4r^2 + 3r + 3$$

$$4r^2 + 3r + 3 - 3r^2 - 9r - 10 = 0$$

$$r^2 - 6r - 7 = 0$$

$$r^2 - 7r + 1r - 7$$

$$r(r+1) - 7(r+1) = 0$$

$$(r-7)(r+1) = 0, \quad \underline{r=7} \quad r=-1$$

$$c) 221505_8 = 12345_r$$

$$= 2 \times 8^5 + 2 \times 8^4 + 1 \times 8^3 + 5 \times 8^2 + 0 \times 8^1 + 5 \times 8^0$$

$$= 65536 + 8192 + 512 + 320 + 0 + 5 = \underline{74565}$$

$$\Rightarrow 12345_r = 74565$$

$$r^4 + 2r^3 + 3r^2 + 4r + 5 = 74565$$

Assuming $r = 16$

$$16^4 + 2 \times 16^3 + 3 \times 16^2 + 4 \times 16 + 5 = 65536 + 8192 + 768 + 64 + 5$$

$$= 74565$$

$$\therefore LHS = RHS$$

$$\therefore \underline{r = 16}$$

$$\Rightarrow 221505_8 = 12345_{(16)}$$

47) Addition

a) $1101_2 + 1101111_2$

$$\begin{array}{r} 1101111 \\ + 1101 \\ \hline 11101100 \end{array}$$

b) $77652_8 + 76543_8$

$$\begin{array}{r} 7^1 7^1 6^1 5^1 2 \\ + 7^1 6^1 5^1 4^1 3 \\ \hline 176415 \end{array}$$

$$\begin{array}{r} 8 \overline{)9} \\ 1 - \end{array}$$

$$\begin{array}{r} 8 \overline{)12} \\ 1 - 4 \end{array}$$

$$\begin{array}{r} 8 \overline{)14} \\ 1 - 6 \end{array}$$

$$\begin{array}{r} 8 \overline{)15} \\ 1 - 7 \end{array}$$

c) $578A_{16} + ABDE_{16}$

$$\begin{array}{r} 5^1 7^1 8^1 A \\ + A^1 B^1 D^1 E^1 \\ \hline 10368 \end{array}$$

$$\begin{array}{r} 16 \overline{)24} \\ 1 - 8 \end{array}$$

$$\begin{array}{r} 16 \overline{)22} \\ 1 - 6 \end{array}$$

$$\begin{array}{r} 16 \overline{)19} \\ 1 - 3 \end{array}$$

$$\begin{array}{r} 16 \overline{)16} \\ 1 - 0 \end{array}$$

d) $234_6 + 315_6$

$$\begin{array}{r} 2^1 3^1 4^1 \\ + 3^1 1^1 5^1 \\ \hline 553 \end{array}$$

$$\begin{array}{r} 6 \overline{)9} \\ 1 - 3 \end{array}$$

5) Binary $\rightarrow$ Gray Code

a) 10101₍₂₎

$$\begin{array}{ccccccc} & & 1 & \oplus & 0 & \oplus & 1 & \oplus & 0 & \oplus & 1 \\ & \downarrow & & & & & & & & & \\ \rightarrow & 1 & & 1 & & 1 & & 1 & & 0 & & 1 \end{array}$$

b) 01110₍₂₎

$$\begin{array}{ccccccc} & & 0 & \oplus & 1 & \oplus & 1 & \oplus & 1 & \oplus & 0 \\ & \downarrow & & & & & & & & & \\ \rightarrow & 0 & & 1 & & 0 & & 0 & & 0 & & 1 \end{array}$$

6) Gray Code $\rightarrow$ Binary

a) 01101_(gray)

$$\begin{array}{ccccccc} 0 & 1 & 1 & 0 & 1 \\ \downarrow \oplus \downarrow & & & & & & \\ 0 & 1 & 0 & 0 & 1 \end{array}$$

b) 10101_(gray)

$$\begin{array}{ccccccc} 1 & 0 & 1 & 0 & 1 \\ \downarrow \oplus \downarrow & & & & & & \\ 1 & 1 & 0 & 0 & 1 \end{array}$$